

Problem

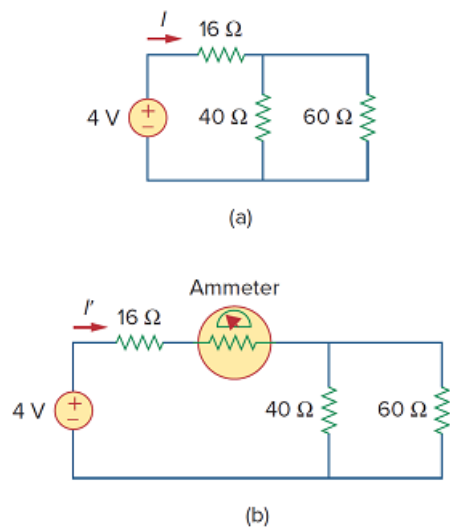
(a) Find the current I in the circuit of Fig. 2.128(a).

(b) An ammeter with an internal resistance of $1\ \Omega$ is inserted in the network to measure I' as shown in Fig. 2.128(b). What is I' ?

(c) Calculate the percent error introduced by the meter as

$$\left| \frac{I - I'}{I} \right| \times 100\%$$

Figure 2.128:



Step-by-step solution

Step 1 of 6

(a)

Consider the following circuit diagram:

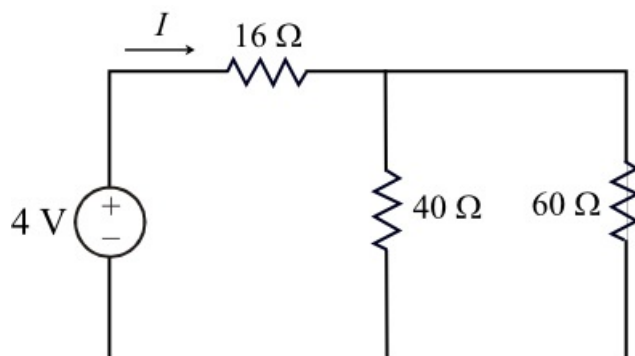


Figure 1

Step 2 of 6

In the Figure 1, the resistors $40\ \Omega$ and $60\ \Omega$ are connected in parallel and resultant is in series with $16\ \Omega$ resistor. Therefore, their equivalent impedance is

$$\begin{aligned}R_{eq} &= (40 \parallel 60) + 16 \\&= \frac{(40)(60)}{40 + 60} + 16 \\&= 24 + 16 \\&= 40\ \Omega\end{aligned}$$

Step 3 of 6

Calculate the current I in the circuit shown in Figure 1.

$$\begin{aligned}I &= \frac{4}{R_{eq}} \\&= \frac{4}{40} \\&= 0.1\ \text{A}\end{aligned}$$

Therefore, the current I in the circuit is 0.1 A.

Step 4 of 6

(b)

Consider the following circuit diagram:

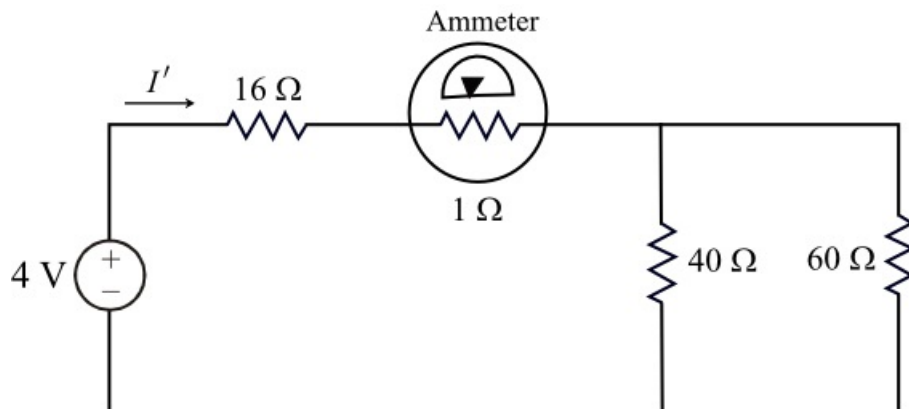


Figure 2

Step 5 of 6

In the Figure 2, the resistors $40\ \Omega$ and $60\ \Omega$ are connected in parallel and resultant is in series with $16\ \Omega$ resistor and $1\ \Omega$ resistor. Therefore, their equivalent impedance is

$$\begin{aligned} R'_{eq} &= (40 \parallel 60) + 16 + 1 \\ &= \frac{(40)(60)}{40 + 60} + 16 + 1 \\ &= 24 + 16 + 1 \\ &= 41\ \Omega \end{aligned}$$

Calculate the current I' in the circuit shown in Figure 2.

$$\begin{aligned} I' &= \frac{4}{R'_{eq}} \\ &= \frac{4}{41} \\ &= 0.09756\ \text{A} \end{aligned}$$

Therefore, the current I' in the circuit is $\boxed{0.09756\ \text{A}}$.

Step 6 of 6

(c)

Calculate the percent error introduced by the meter.

$$\left| \frac{I - I'}{I} \right| \times 100 \dots\dots (1)$$

Substitute $0.1\ \text{A}$ for I and $0.09756\ \text{A}$ for I' in equation (1).

$$\begin{aligned} \left| \frac{I - I'}{I} \right| \times 100 &= \left| \frac{0.1 - 0.09756}{0.1} \right| \times 100 \\ &= \left| \frac{0.00244}{0.1} \right| \times 100 \\ &= 0.0244 \times 100 \\ &= 2.44\% \end{aligned}$$

Therefore, the percent error introduced by the meter is $\boxed{2.44\%}$.